Green IoT: Optimising Wireless Sensor Networks

A comprehensive architecture integrating solar prediction, intelligent scheduling, and energy-aware routing for robust, long-term wireless sensor networks.

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CHAPTER 1

Integrated Architecture: A Synergistic Approach

Our proposed framework combines the strengths of three key research papers, creating a symbiotic relationship where each component enhances the others for superior performance in Green IoT environments.

1

Solar Prediction

High-accuracy forecasting

2

Energy Scheduling

Optimised node activity

3

Adaptive Routing

Load-balanced data paths

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Paper Focus: Core Techniques and Benefits

Each foundational paper brings a critical component to our integrated Green IoT solution, addressing specific challenges in energy management and network longevity.

Prediction: Modified Pro-Energy

Utilises high-accuracy solar forecasting with a self-correction mechanism (ρ) to minimise errors, ensuring reliable energy intake estimates.

Routing: ERAS Strategy

Employs energy-aware multi-hop routing that balances network load and extends network lifetime by considering node energy levels (FND, HND, LND).

Scheduling: ESTC Algorithm

A target coverage-based approach that rotates nodes between active and sleep modes using non-disjoint sets, optimising coverage and energy conservation.

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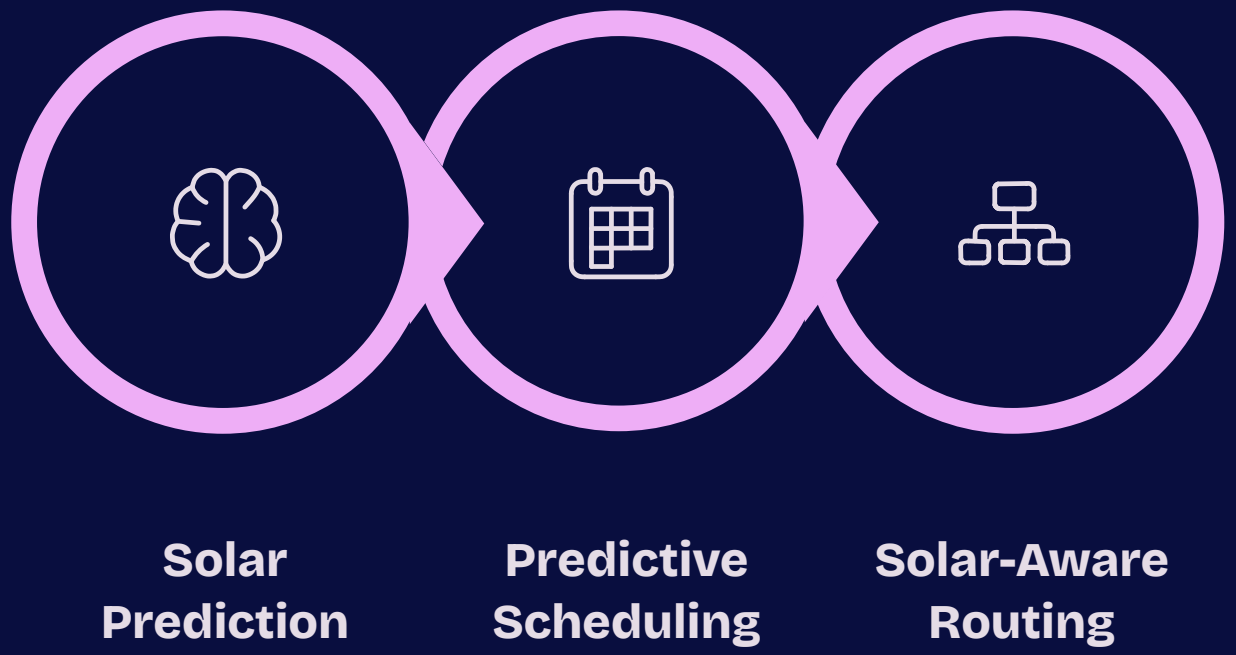
CHAPTER 2

Integrated Workflow Proposal

Our innovative approach leverages solar prediction as a central parameter, informing and optimising both routing and scheduling within a cohesive Green IoT framework.



This workflow ensures that energy management is proactive and adaptive, significantly boosting network resilience and efficiency.

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Step 1: Solar Prediction as the 'Brain'

The modified Pro-Energy model acts as the central intelligence, providing critical energy forecasts to guide subsequent network operations.

Forecasting Energy Intake

Every node continuously forecasts its expected solar energy intake (E_{solar}) for the upcoming time slot. This projection is crucial for informed decision-making.

Predicted Residual Energy (PRE)

This forecasted E_{solar} is directly fed into both the scheduling and routing modules as a 'Predicted Residual Energy' metric, ensuring all decisions are energy-aware.

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Step 2: Predictive Scheduling with ESTC

The ESTC algorithm is enhanced with predictive capabilities, moving beyond standard on/off switching to intelligent, energy-neutral activation.

Prediction-based Activation

If low sunlight is predicted (e.g., cloudy days), ESTC aggressively increases sleep cycles for nodes in t_{max} sets, prioritising energy for critical nodes ($t_{critical}$).

Energy Neutrality

Nodes only activate if their predicted harvested energy meets or exceeds their expected consumption for that slot, preventing premature energy depletion.

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Step 3: Solar-Aware Routing with ERAS

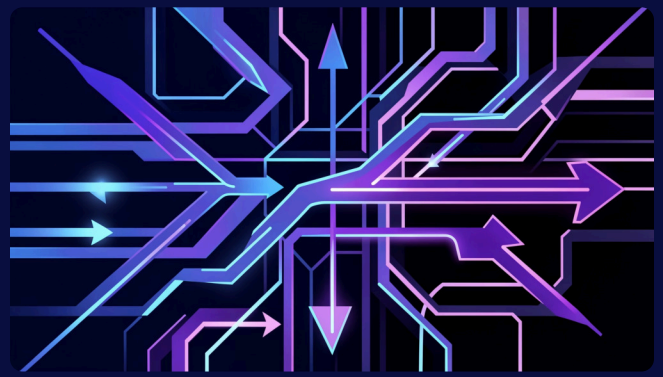
The ERAS algorithm is modified to prioritise data paths based on future solar energy availability, ensuring robust and efficient data delivery.

Harvesting-First Priority

When selecting a next-hop (PNH), ERAS considers not only current residual energy but also 'Predicted Total Energy' (Current + Predicted Solar).

Energy-Rich Paths

Data is routed through nodes predicted to have a 'sunny' forecast, strategically avoiding nodes projected to be in 'shadow' or with low incoming energy.



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CHAPTER 3

Enhancing with Artificial Neural Networks (ANN)

An Artificial Neural Network can provide a significant upgrade to the linear models, enabling more complex pattern recognition and refined decision-making in Green IoT.

High-Precision Prediction

Replacing Pro-Energy for superior solar irradiance forecasts.

Adaptive Routing Decisions

Optimising path selection based on historical success data.

Anomaly Detection

Identifying faulty or obstructed nodes for rerouting.

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ANN for High-Precision Prediction

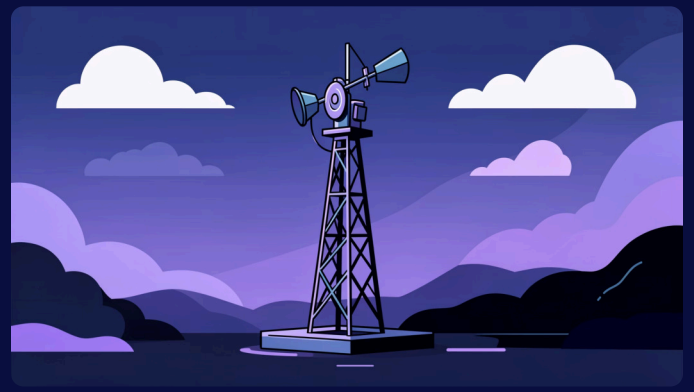
While Pro-Energy is efficient, an ANN can drastically improve solar forecasting accuracy, especially in dynamic weather conditions.

Complex Pattern Recognition

ANNs can better handle rapid weather fluctuations by incorporating multiple inputs such as temperature, humidity, and wind speed.

Superior Accuracy

Achieves much higher accuracy in predicting solar irradiance compared to simpler profile matching techniques.



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ANN for Adaptive Routing and Anomaly Detection

Beyond prediction, ANNs offer sophisticated capabilities for dynamic routing adjustments and proactive fault identification within the network.

Adaptive Routing Decisions

An ANN can be trained on historical routing successes to select the 'Best Path', weighing factors like distance, current battery levels, and predicted solar intake.

Anomaly Detection

A lightweight ANN can detect 'faulty' nodes or coverage holes by flagging significant discrepancies between predicted and actual energy harvest. This enables ESTC to reroute coverage sets promptly.

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Should we use ANN for Routing and coverage? We are aiming to improve this

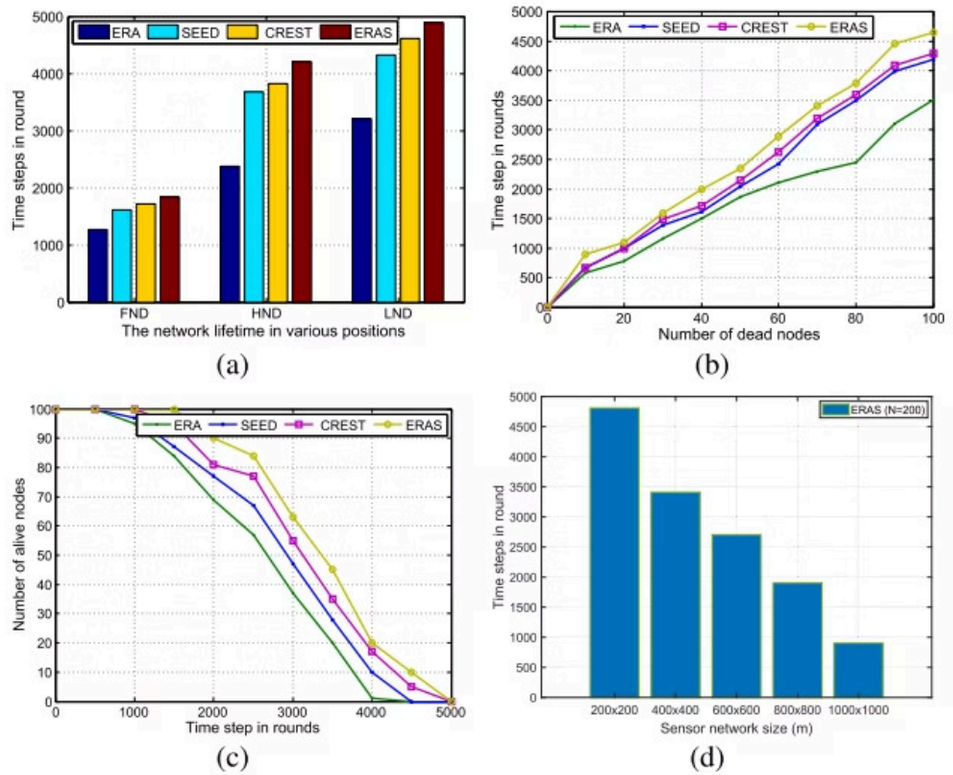

Fig. 5. Network lifetime analysis. (a) Network lifetime at various positions as FND, HND, and LND within rounds. (b) Network lifetime of dead nodes with standard techniques. (c) Number of alive nodes in standard algorithms. (d) Performance of ERAS strategy.

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🛰️ Optimized Integrated Framework: Coverage-Driven Execution

Phase 1: Predictive Energy Profiling (The "Budget" Phase)

Before any action, nodes determine their potential to be leaders.

- **Solar Prediction**: Each node uses the **Modified Pro-Energy** model (or **ANN**) to estimate incoming solar energy **E_solar**.
- **Energy Status**: Nodes calculate their **Residual Energy E_resi**.
- **Health Score Calculation**: An ANN combines **E_resi** and predicted **E_solar** to output a "Health Score," signaling how long the node can stay active without dying.

Phase 2: Target Coverage Activation (The "Sensing" Phase)

"Minimum Active Set."

- **Cover Set Selection**: The system identifies which **non-disjoint cover sets C_i** are needed to cover all **Target Areas (TAs)**.
- **Node Activation**: Only the nodes required for the current cover set are turned **ON**; the rest are put into **Sleep Mode** to maximize solar harvesting.
- **Critical Identification**: Nodes covering **t_critical** (targets with the fewest covering nodes) are prioritized for immediate activation.

Phase 3: Cluster Election & Setup (The "Organization" Phase)

Now that we know which nodes are "ON," we apply the **Phase 1 (Clustering)** logic to these active nodes.

- **CH Selection**: Among the active nodes, those with the highest "Health Scores" from Phase 1 elect themselves as **Cluster Heads (CH)**.
- **Role Assignment**: These leaders become responsible for the active **Time Steps** and coordination.
- **Intra-Cluster TDMA**: The CH assigns sensing slots to its active members, ensuring they don't all transmit at once, which saves radio energy.

Phase 4: Distance-Aware Routing (The "Delivery" Phase)

This phase ensures the sensed data actually reaches the destination.

- **Data Aggregation**: CHs compress data from their members to minimize the payload size.
- **Next-Hop (PNH) Selection**: Using the **ERAS** logic, the CH finds the best next-hop neighbor based on the **distance to the Base Station** and the neighbor's **predicted energy**.
- **Multi-Hop Transmission**: If the distance **d** >= **r_c ,** data is relayed through intermediate active nodes to avoid the high power cost of long-range transmission.

Phase 5: Learning & Self-Correction (The "Optimization" Phase)

- **Error Update**: Nodes compare the actual energy harvested during Phase 2 with the prediction from Phase 1.
- **Model Refinement**: This error **rho** is used to train the ANN, making the "Health Score" in the next round even more accurate.

Why this order (Phase 2 before Phase 3) is better:

1. **Reduces Overhead**: You aren't wasting energy forming clusters with nodes that don't need to be awake for target coverage.
2. **Prolongs Lifetime**: Nodes that aren't part of the "Active Set" get a 100% duty cycle for **Solar Harvesting**, allowing them to build up energy for future rounds.
3. **Optimized CHs**: All elected Cluster Heads are guaranteed to be "useful" because they are already covering a target.

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